

Deep Learning Guided Discovery and FEP Validation of NOD2 Binders for the R702W Crohn's Disease Variant

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INTRODUCTION

NOD2 mutations are among the strongest genetic risk factors for Crohn's disease, increasing risk up to **40-fold**.

The **R702W variant** is the most prevalent, affecting ~10% of Crohn's patients.

No drugs currently target NOD2 — a significant unmet need.

RESEARCH OBJECTIVES

- Discover compounds that bind NOD2
- Validate binding using gold-standard FEP
- Characterize how R702W affects binding

NOD2 PROTEIN STRUCTURE

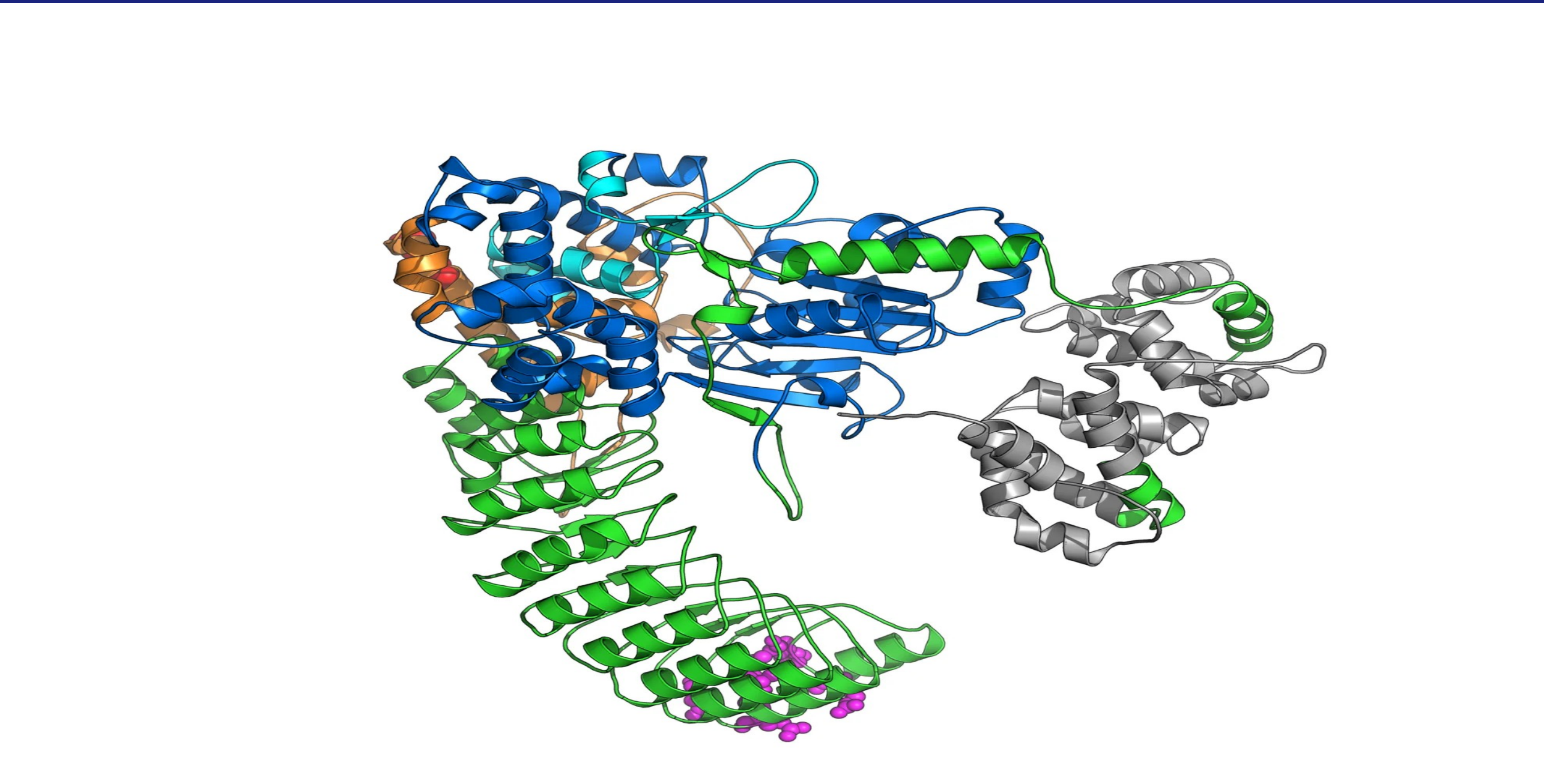
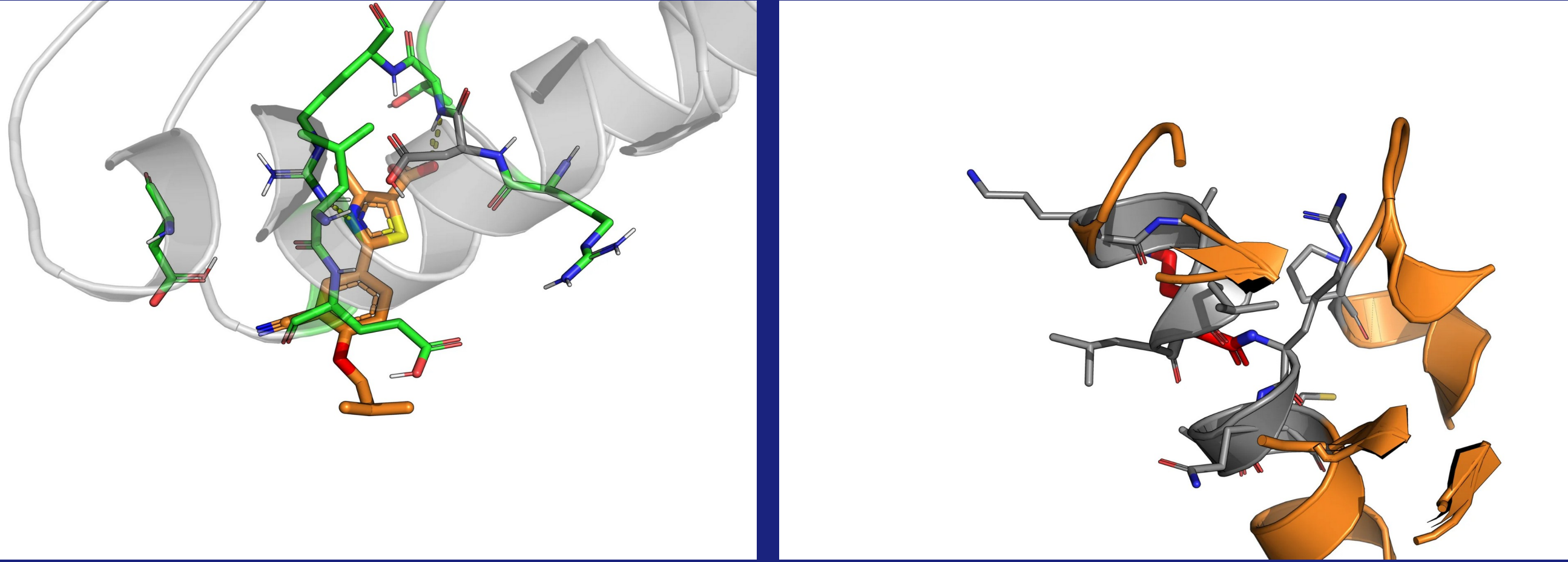


Fig 1. NOD2 domains: CARD (gray), NBD (blue), HD2 (orange, R702W), LRR (green, binding). Created by Finalist using PyMOL.

BINDING POCKET & R702W SITE



Key residues: GLU1008, ARG1037, ASP1011

Fig 2-3. Binding pocket (left), R702W site 80Å from pocket (right). Created by Finalist using PyMOL.

COMPUTATIONAL PIPELINE

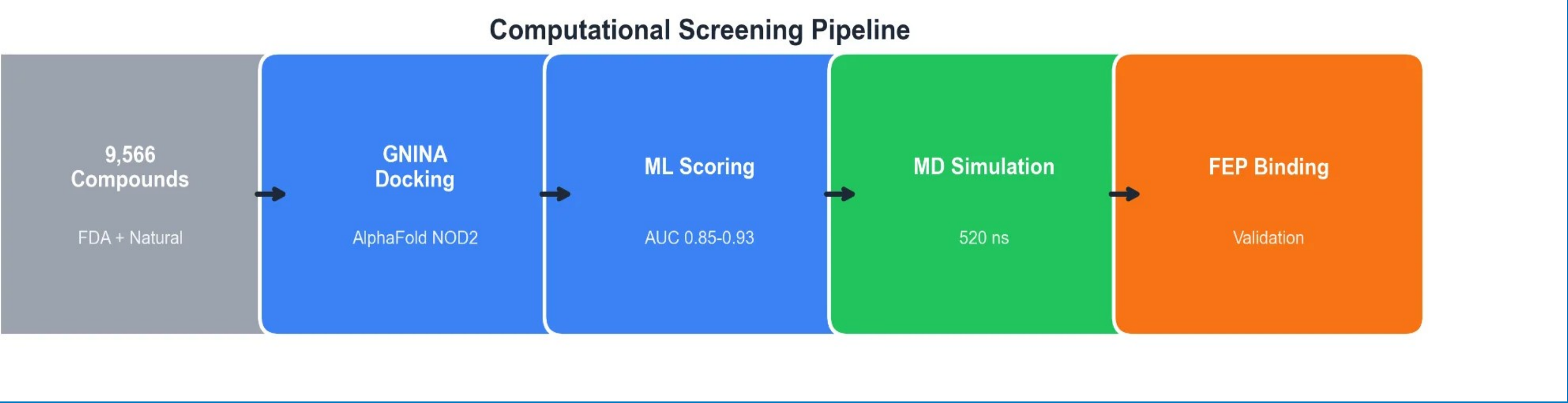


Fig 4. Screening pipeline: 9,566 compounds → GNINA docking → ML scoring (AUC 0.85-0.93) → MD validation → FEP binding. Created by Finalist.

MATERIALS & METHODS

Compound Screening

- 9,566 compounds screened (FDA + natural products)
- GNINA CNN-based docking
- NOD2-Scout ML classifier (AUC 0.85-0.93, scaffold CV)

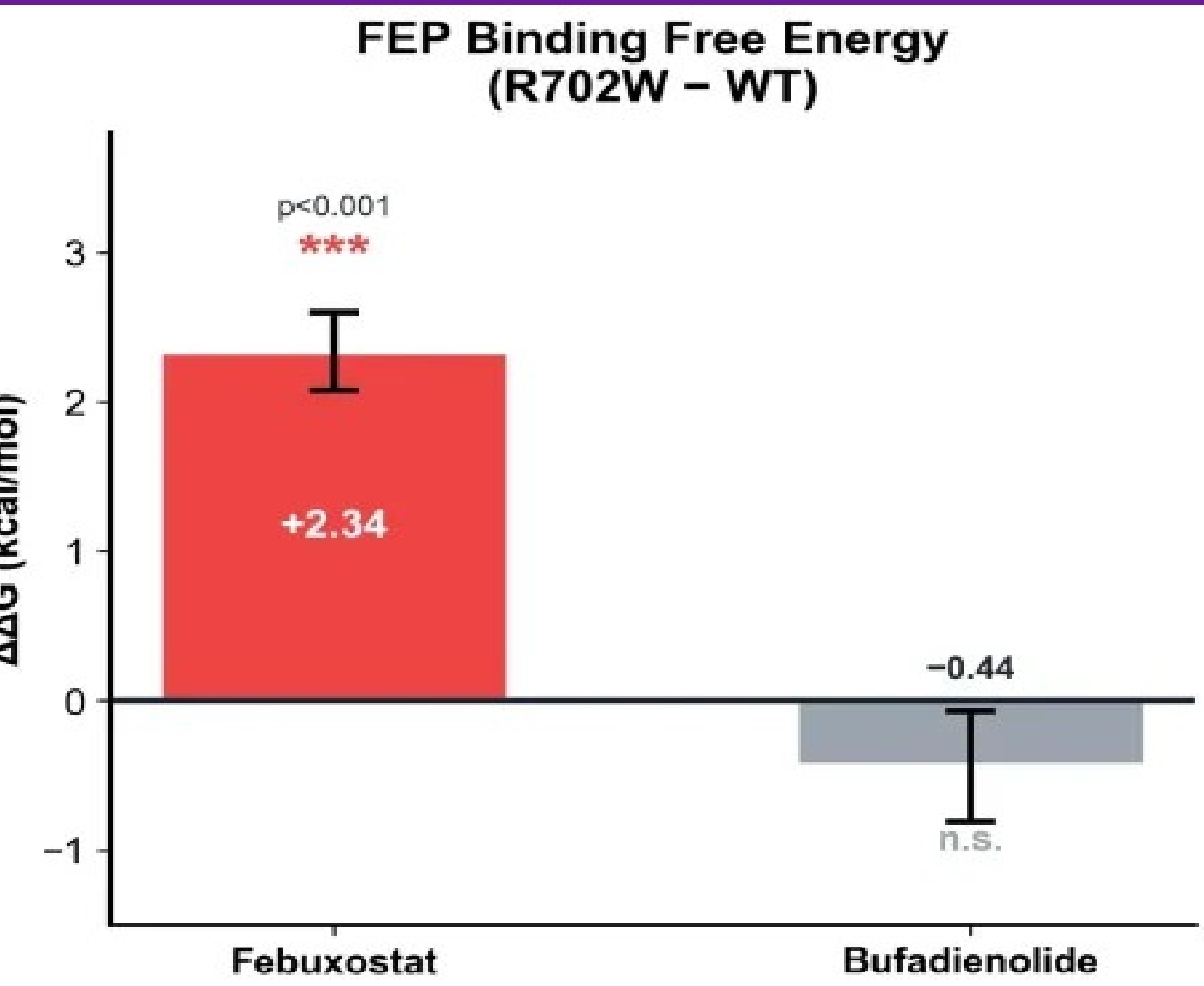
MD Simulation

- OpenMM 8.0
- AMBER14 + OpenFF 2.1.0
- TIP3P-FB water, 310 K
- 520 ns total simulation
- WT apo (n=4), R702W (n=3)

FEP Calculations

- Gold-standard binding free energy
- 120 lambda windows (80 complex + 40 solvent)
- MBAR analysis (pymbar)

★ KEY RESULT: FEP BINDING FREE ENERGY ★



Febuxostat

$\Delta\Delta G = +2.34$ kcal/mol

50x weaker binding in R702W

$p < 0.001$ • 8σ significance ***

Bufadienolide

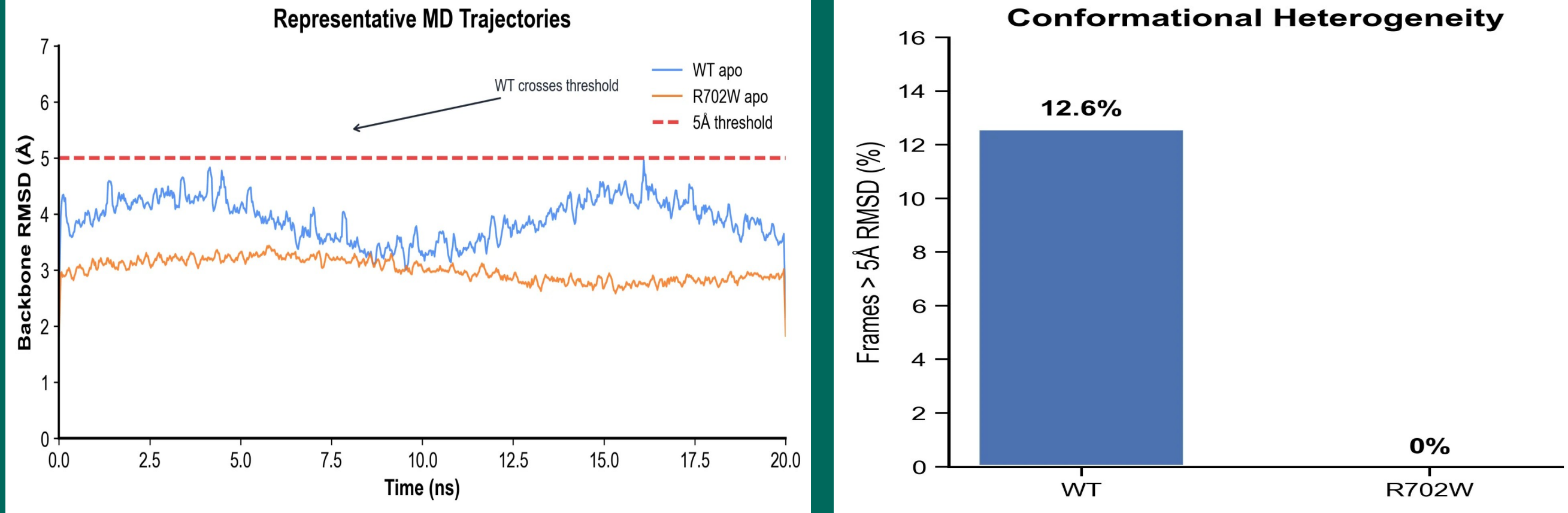
$\Delta\Delta G = -0.44$ kcal/mol • No significant change (NS)

ΔG bind (WT): -10.36 ± 0.18 kcal/mol | ΔG bind (R702W): -8.02 ± 0.19 kcal/mol

KEY INSIGHT: Ligand-dependent binding differences between WT and R702W
Febuxostat is MUTATION-SENSITIVE | Bufadienolide is UNAFFECTED

Fig 5. FEP $\Delta\Delta G$ (R702W - WT). Positive $\Delta\Delta G$ indicates weaker binding in mutant. Created by Finalist using Python/matplotlib.

APO MD DYNAMICS



REDUCED CONFORMATIONAL SAMPLING IN R702W

WT: 12.6% frames >5Å RMSD | R702W: 0% frames >5Å RMSD

Consistent with mutation-sensitive binding observed in FEP

Fig 6-7. RMSD trajectories and conformational heterogeneity. WT shows greater sampling. Created by Finalist.

RESULTS SUMMARY

FEP Binding Validation:

- Febuxostat: ΔG (WT) = -10.36 , ΔG (R702W) = -8.02
- Bufadienolide: ΔG (WT) = -15.22 , ΔG (R702W) = -15.66

Key Binding Residues:

GLU1008, ARG1037, ASP1011

Allosteric Mechanism:

R702W site is 80Å from binding pocket

CONCLUSIONS

- Built deep learning pipeline screening 9,566 compounds
- FEP validates mutation-sensitive binding for Febuxostat (50x weaker in R702W, $p < 0.001$)
- R702W shows reduced conformational sampling (0% vs 12.6% frames >5Å)
- Allosteric mechanism: mutation 80Å from binding pocket

FUTURE WORK

- SPR/ITC experimental validation
- HDX-MS confirmation
- Test G908R, L1007fs variants

REFERENCES

- Hugot JP et al. (2001) Nature 411:599
- Jumper J et al. (2021) Nature 596:583
- McNutt AT et al. (2021) JCI 131:2522